UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. How many times does a while loop execute when its condition is false initially?
   A. One
   B. Zero
   C. Until the user stops it
   D. Two

2. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. p
   B. t
   C. i
   D. n

3. Which tool represents an algorithm using standard graphical symbols?
   A. Pseudocode
   B. Debugger
   C. Compiler
   D. Flowchart

4. What is displayed by the following code?

```
int a[4] = {3, 6, 9, 12};
cout << a[1] + a[3];
```

   A. 9
   B. 12
   C. 18
   D. 15

5. A computer system's hardware includes which of the following?
   i. input devices
   ii. output devices
   iii. central processing unit
   iv. main memory
   v. secondary storage
   A. i, ii, iii, v
   B. i, ii, iii, iv, v
   C. i, ii, iii, iv
   D. i, iii, iv

6. Which keyword prevents a variable's value from being changed after initialisation?
   A. const
   B. static
   C. void
   D. break

7. Which statement is most suitable for selecting among several constant cases?
   A. do...while

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B. switch
C. for
D. while

8. Which declaration creates a floating-point variable named price?
A. float price;
B. int price;
C. char price;
D. bool price;

9. Which of the following are stream operators?
i. <<
ii. >>
iii. ==
iv. =
A. i and iii
B. i and ii
C. iii and iv
D. ii and iv

10. Which statement immediately exits a loop?
A. break
B. continue
C. cin
D. return 0

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(8 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(5 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(5 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
    s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A catering company charges the meal prices shown in Table 1.

Table 4. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

END OF QUESTION PAPER

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